

Hybrid performance across environments

Satellite-informed forecasts for better hybrid selection across environments.

2022 trials in Nebraska and Iowa: **84 hybrids, five locations, 2,131 labeled plots.**

Observed shortlist for the next replicated trials

All five candidates exceed the median at every site. Replicates are averaged within treatment, treatment percentiles within site, and sites equally. These are observed harvest findings, not forecasts.

Hybrid	Avg. percentile	Worst site	Plots
HOEGEMEYER 8065RR	93.4	84.3	25
SYNGENTA NK0760-3111	87.5	74.7	64
HOEGEMEYER 7089 AMXT	86.5	67.7	26
PIONEER P0589 AMXT	85.0	67.5	26
PHW52 X PHN82	81.7	58.9	26

Some treatment cells have only one replicate. Shortlist eligibility requires above-median results at four or more sites.

Nitrogen contrasts depend on the site

Observed 150-minus-75 lb nitrogen/acre differences, matching 84 hybrids within each of four sites:

Ames	Crawfordsville	Lincoln	Scottsbluff
-23.6 bu/ac	+15.4 bu/ac	+10.9 bu/ac	+29.2 bu/ac

The equal-site mean is +8.0 bu/ac across 336 matched hybrid-site pairs. Field placement may contribute. These are descriptive contrasts, not optimal-rate recommendations.

Forecast evidence at unseen locations

Day-90 model	MAE (bu/ac)	Top-10 overlap per site
Training-mean baseline	55.1	Cannot rank hybrids
CatBoost with imagery	44.0	4.6 of 10
Ridge with imagery	39.4	5.0 of 10

Ridge reduces MAE by 28.4% versus the mean baseline. Random top-10 selection has an expected overlap of 1.19. Ridge is exploratory 2022 development with tuning inside training folds; forecasts remain imperfect.

Decision and next test

Prioritize the observed shortlist for replicated trials. Validate a frozen forecasting pipeline on untouched 2023 data before claiming cross-season reliability.

Scope: five locations in one season. No causal irrigation effect, guaranteed yield improvement, or financial return is established.